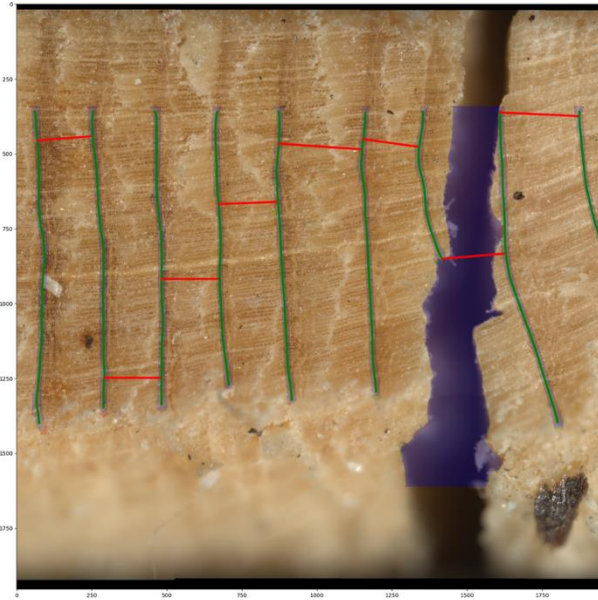


Automated ring-width measurements from images

Miroslav Poláček



Quiz

- Have you ever used terminal?
- Have you ever used python?
- Have you ever worked with neural networks?
- What do you use for measuring cores and crossdating?

Some useful terminal commands

- **ls** – list content of a directory
- **pwd** – print where you are now
- **cd /mnt/c** –change directory
 - **cd ../..** – go 2 directories up
- **mkdir** – make new directory
- **rm** – delete file
 - **rm -r** – delete directory and everything in it
- **sh file.sh** – will run bash script in file.sh
- **python file.py** – will run python code in file.py

Instalations and set up

- Wsl – only Windows users to have Linux
- Miniconda – all operating systems
- Install Conda environment
 - **conda create -n TRACEW2 python==3.12.9**
 - **conda activate TRACEW2**
 - **pip install ultralytics lxml shapely scikit-image ujson numba openpyxl 'numpy<2.0.0'**

Instalations and set up

- Clone the repository
 - **cd /mnt/c/...** where you want – for Windows users
 - **cd** where you want – Mac and Linux users
 - **git clone -b TRACE2025W2 <https://github.com/Gregor-Mendel-Institute/TRG-ImageProcessing.git>**
 - **cd TRG-ImageProcessing**
 - **git lfs pull**
- Test run
 - **cd CoreProcessingPipelineScripts/CNN/bash_scripts**
 - **sh test_run.sh**
 - Can take around 1 or 2 min

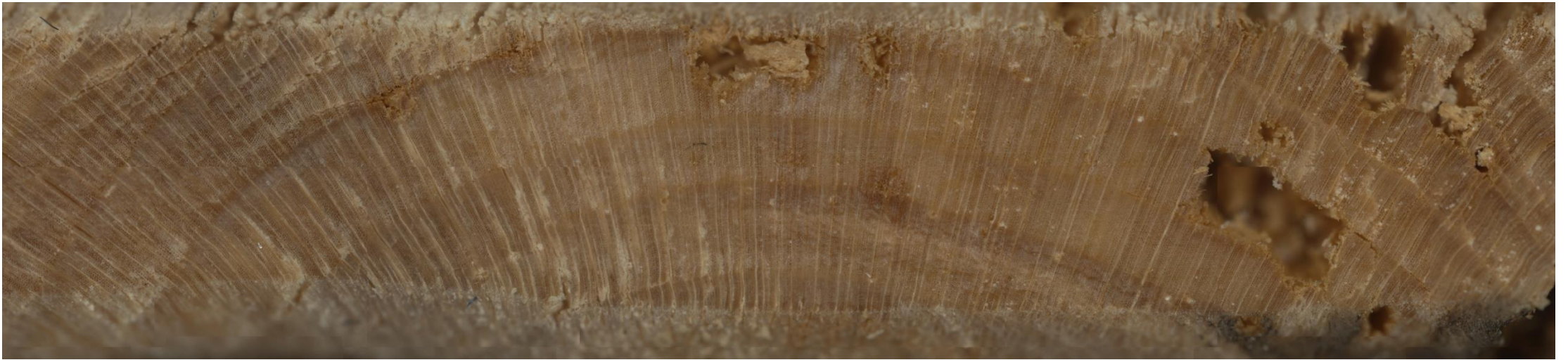
Why to automate?

- Reduce manual work
- Reduce human error

Core measurement automation

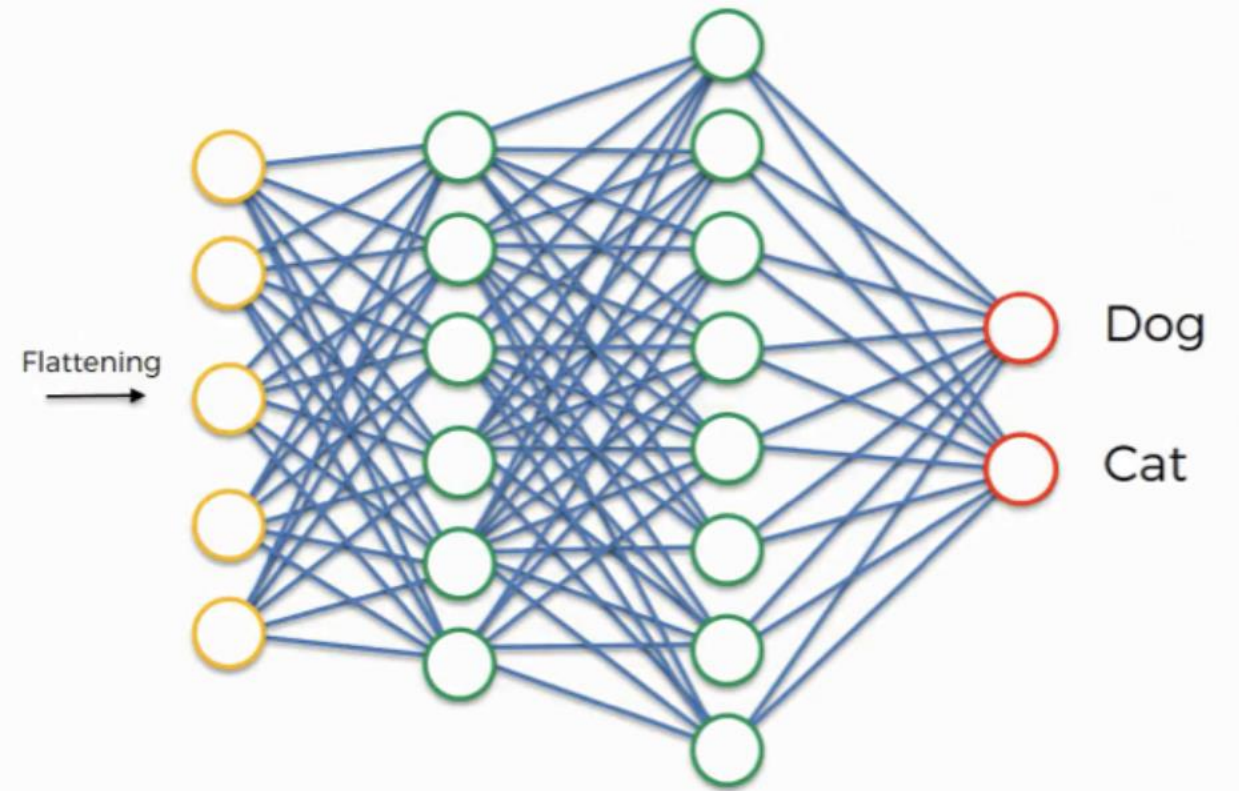


Core measurement automation



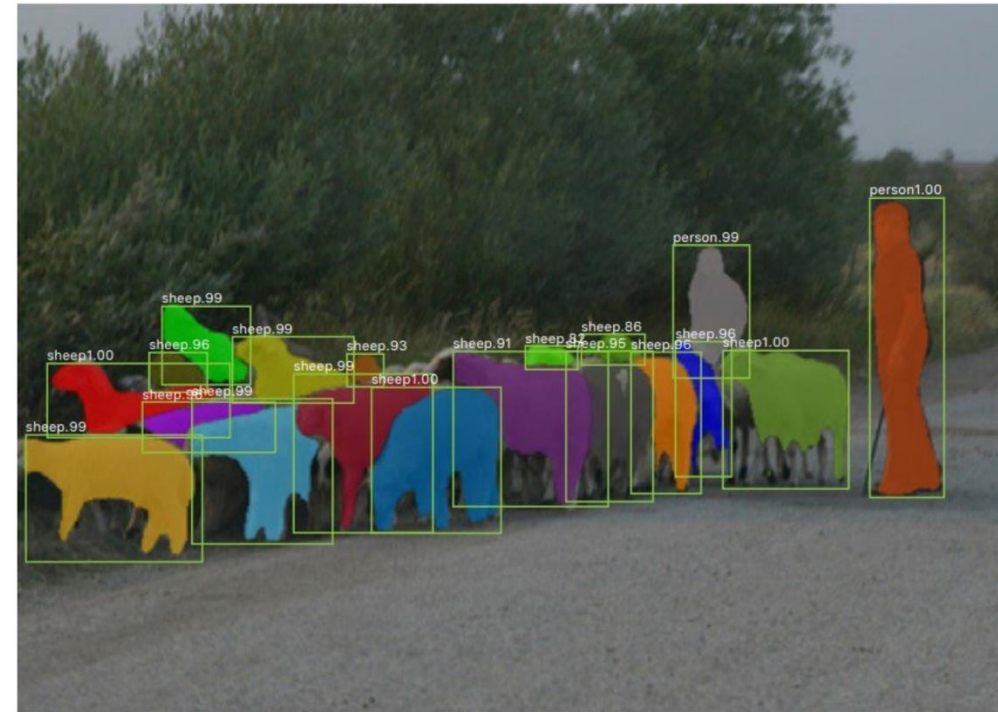
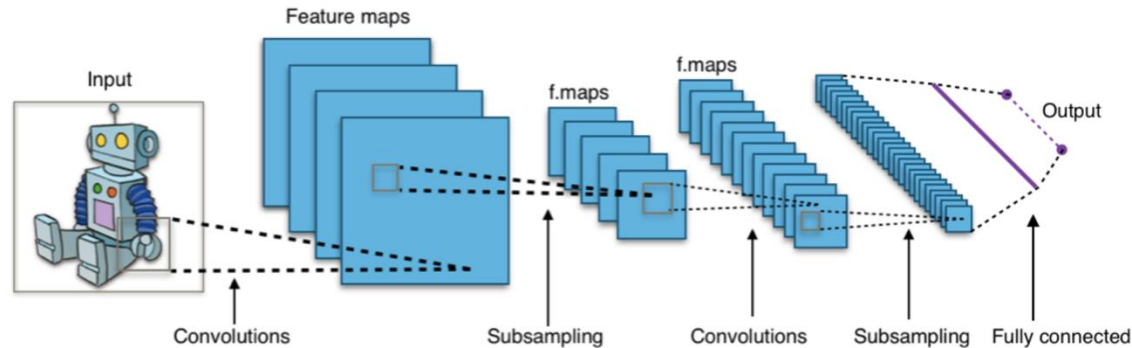
Neural networks

- Machine learning algorithms
- Learn complex features from the data
- Need big datasets to learn from
- Computationally heavy
 - GPUs
- Parameters - weights



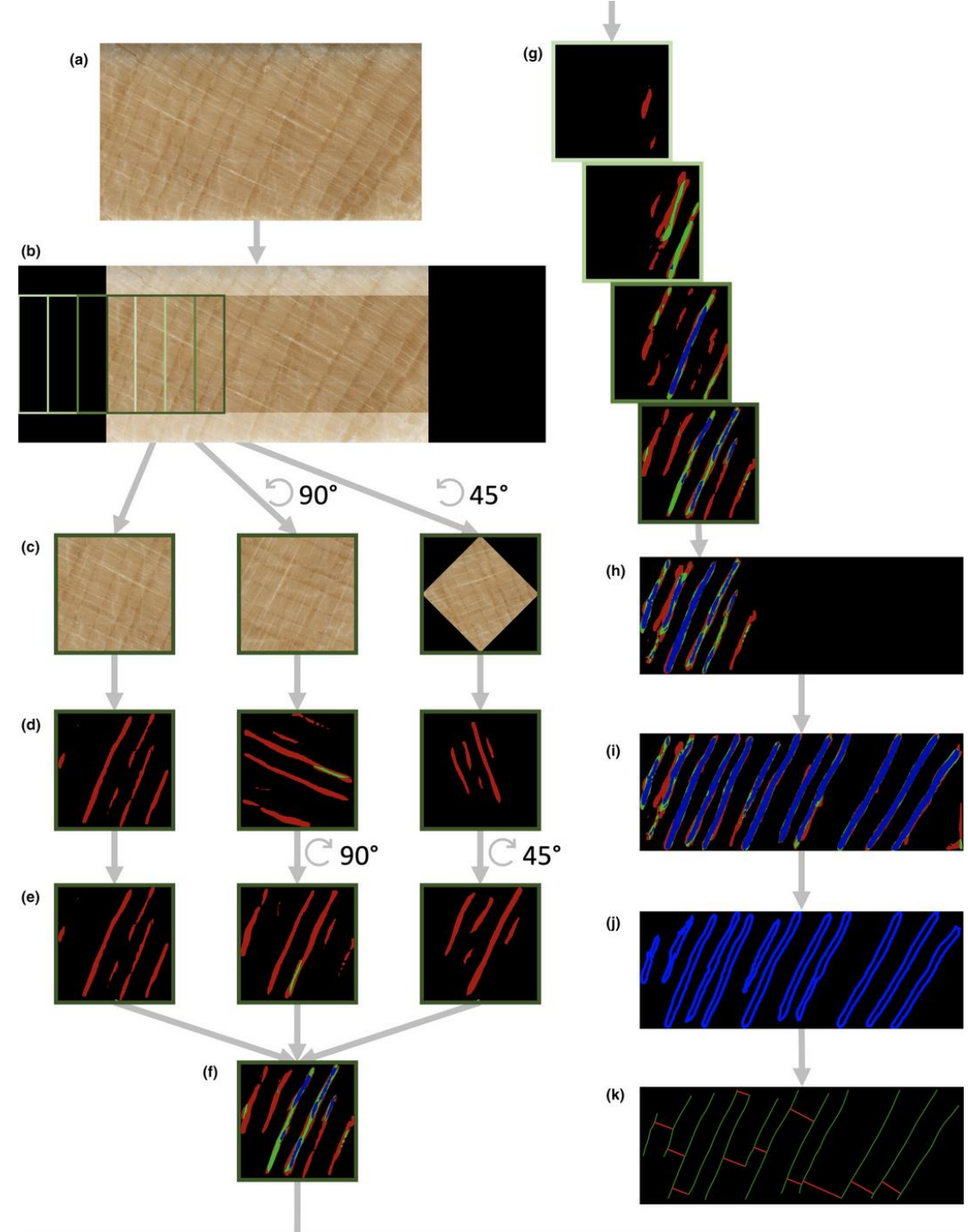
Convolutional neural networks

- Computer vision
 - Face detection, self driving cars
- Classification
- Object detection
- Segmentation



Our implementation

- YOLOv8
- Complex postprocessing



How to run it?

- Directly python code in virtual environment
 - Anaconda, **Miniconda**, venv...
- Container
 - Apptainer, Docker

How to run it?

- **cd CoreProcessingPipelineScripts/CNN/bash_scripts**
- **sh inference_laptop.sh**

```
time python ../processing/processing.py \  
  --dpi=13039 \  
  --run_ID=Debugging_centerlines \  
  --input=../training/sample_dataset/val/ \  
  --weights=../weights/best10px1000eAugEnlargedDataset.pt \  
  --output_folder=../output \  
  --cropUpandDown=0.17 \  
  --sliding_window_overlap=0.75 \  
  --cracks=True \  
  --debug=True \  
  --print_detections=True\
```

Arguments important for inference

--dpi=13039

--run_ID=inference_run0

--input=../training/sample_dataset/val

--weights=../weights/best10px1000eAugEnlargedDataset.pt

--output_folder=../output

Directory structure of the app

Arguments important for inference

Generally useful for debugging, might slow it down

`--debug=True`

`--print_detections=True`

If you wish to detect cracks

`--cracks=True`

Arguments important for inference

--cropUpandDown=0.17

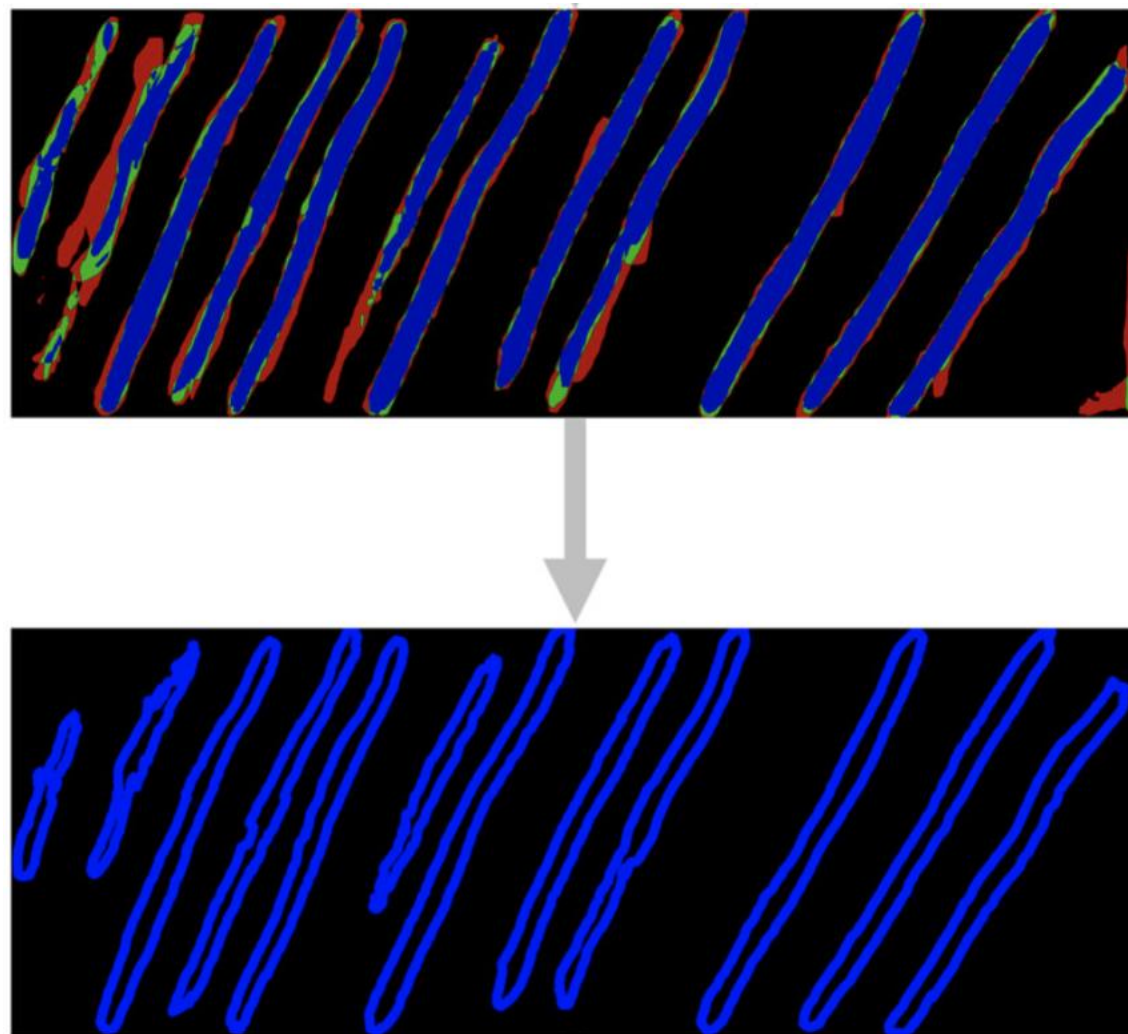
--sliding_window_overlap=0.75

--n_detection_rows=1



Arguments important for inference

--min_mask_overlap=3



Let's run it

- **sh inference_laptop.sh**

Check results

Now is your turn to play

- Modify these arguments to increase the speed while maintaining accuracy

--debug=True

--cracks=True

--cropUpandDown=0.17

--sliding_window_overlap=0.75

--n_detection_rows=1

--min_mask_overlap=3

--run_ID=inference_run0

real 0m56.889s

user 2m2.935s

sys 0m12.306s